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American Football

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Performance Demands of American Football

In this chapter, we introduce the sport of American football (AmF) and discuss the various factors of importance for players and the relative contribution that physical performance may account for in the overall quest to win games.

Sport Description and Factors of Winning

American football (AmF), referred to as *gridiron* in some circles, is a full-contact, team-based sport played predominantly in North America. While Canada hosts a similar four-down version of the game (CFL), the main competition is known as the National Football League (NFL), made up of 32 teams interspersed throughout major cities of the United States, and is currently America's most popular sport. The NFL's final game of the year to decide the overall competition winner is famously known as the Super Bowl, and it is the largest shared event in the United States (7, 18). The 2017 NFL Super Bowl Championship was viewed by approximately 111.9 million people, placing it fourth among the most-viewed television programs in U.S. history (25). The game is played on a rectangle-shaped field (120 yd

long $\times$ 53.3 yd wide) with the last 10 yd at each end of the length of the field allocated as the scoring end zone (figure 20.1). AmF is a unique football code with respect to the fact that it has multiple specialized playing positions that are not interchangeable. Due to the specialized nature of each playing position, professional teams can have a preseason roster of up to 90 players, which is reduced to a 53-man squad once the teams are in-season.

AmF is contested between 11 players on each team (active at any one time) over four, 15 min quarters, separated by a 12 min halftime, with 2 min breaks between first and second and third and fourth quarters (15). Three specific units make up the overall AmF team: a defensive unit, an offensive unit, and a special teams unit. A specific subset of player positions are typically classified into eight groups as follows: defensive backs (DB), defensive linemen (DL), linebackers (LB), offensive linemen (OL), quarterback (QB), running back (RB), tight end (TE), wide receiver (WR), and kickers and punters (38). Each position has distinct physiological demands and activity profiles.

The offensive unit's goal is to advance down the field via running or passing the ball and has four plays (known as downs) to do so. If the offense progresses 10 yd or more in the four plays, they are awarded a new set of four plays. If unsuccessful,

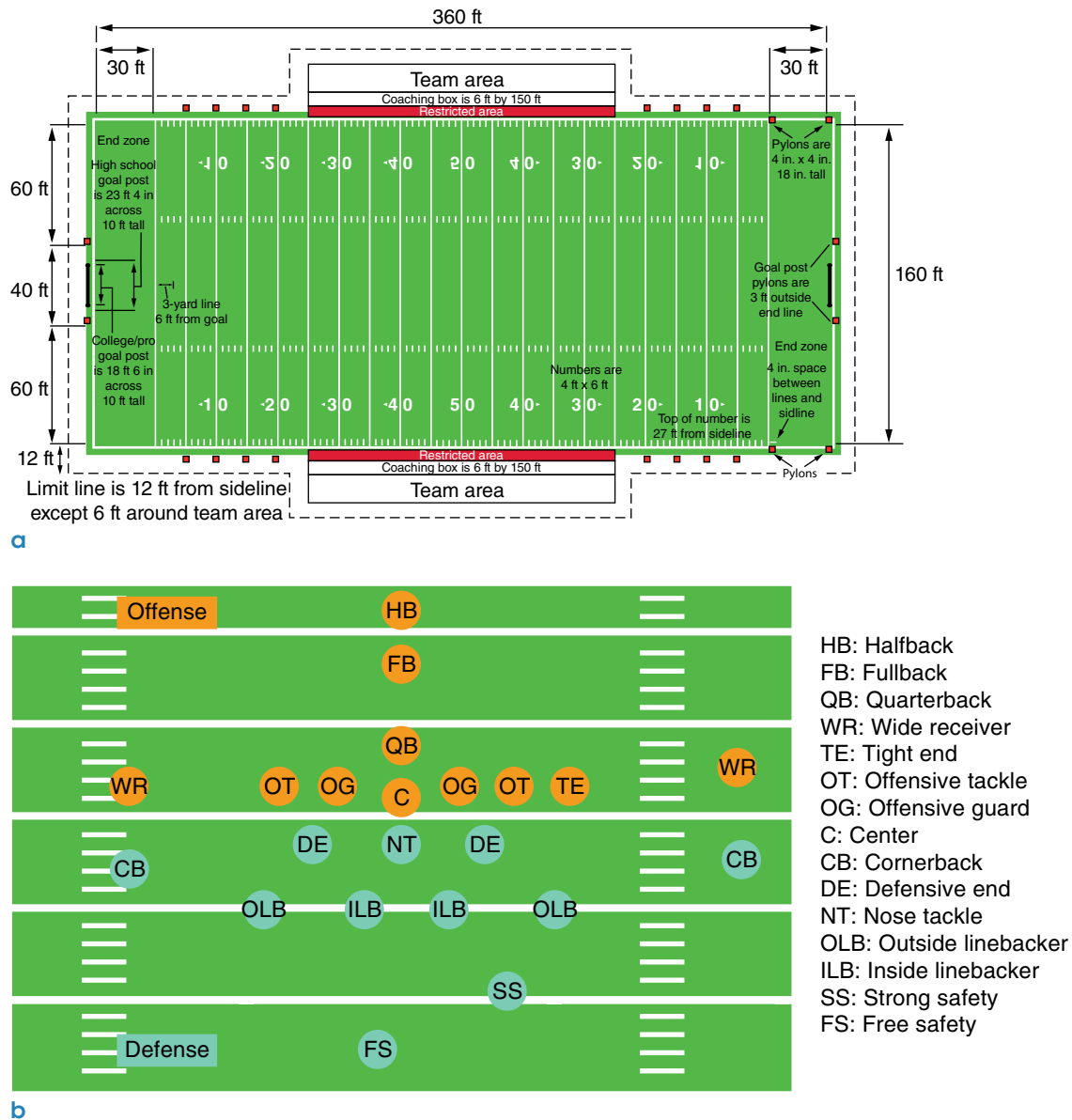


Figure 20.1 (a) American football (AmF) playing field dimensions and (b) playing positions.

then the ball and offensive play are surrendered to the other team by punting. There are two main ways to score points. A touchdown, in which the team progressively moves the ball from one end of the field to their opponent's end zone, earns the team 6 points, with the opportunity to earn an extra point by kicking the ball through the goal posts or try for a 2-point conversion by passing or running the ball into the end zone. The aim of the defensive unit is to protect the length of the field via negating the opposition advancement to score, while a special-teams player's role is to cover situations such as kickoffs, punt returns, and extra point attempts (2, 3).

Relative Contribution of Physical Performance

Despite the popularity of AmF throughout the United States at the high school, college, and professional levels, there is limited research outlining the modern physiological demands of training or games (15, 22, 30). The NFL has only recently shared privately to teams the details of their in-game player tracking via radio-frequency identification (RFID). Despite this, there is to our knowledge no peer-reviewed validation of the system in games for identification of physical demands and capacities of player posi-

tions. To the best of our knowledge, in fact, there are only two published studies that quantify game- or training-based external load in elite AmF players using global positioning systems (GPS) and accelerometers (38-41). Despite this, in-game statistics (yards covered, number of tackles, plays completed, etc.) have been reported in the NFL since 1932 (29), data that may assist practitioners to quantify game demands. This section combines the limited level of empirically based knowledge in AmF with our practical experience working in the sport.

American football players typically have well-established physical profiles of speed, power, strength, and agility (figure 20.2). Successful performers complement these physical characteristics with technical skills and tactical awareness to repeatedly carry out plays successfully during a game while adapting to changes in the environment, e.g., their opposition. The AmF research literature suggests that physical performance characteristics can discriminate between position, playing ability, draft success, and level of play (9, 32, 33, 35, 42). Unlike many other field-based football codes, AmF is often not free-flowing and sees frequent stoppages in play. Each high-intensity play lasts ~5 to

7 s, with recovery times often lasting ~25 to 40 s over a 60 min game, including frequent collisions from blocking or tackling (8, 10, 20). The larger linemen player positions require positional skill and technique plus eccentric, isometric, and concentric strength. The LB, DB, TE, RB, and QB players typically possess certain body mass and speed characteristics that allow them to run and change direction while withstanding high-impact collisions (39). The QB is the key player dictating the orchestration and structure of the team's offense. QBs are required to have high football intellect, skill, and agility plus the ability to read the play quickly and execute decisions, which often requires throwing the football with pinpoint accuracy. Athletes in AmF require the ability to complete repeated sprints (RS) and repeated high-intensity efforts (RHIE) involving accelerations and decelerations combined with tackling and impacts. Data collated from the 16-game 2017 NFL season showed that there were 63 ± 3 (mean $\pm$ SD) scrimmage plays per game, with 5.3 ± 0.5 yd covered per play (29). Certain styles of play, tactical issues, and timeouts may demand athletes to play periods of the game at a higher tempo, which in turn places additional demands on the aerobic system (10).



Figure 20.2 The position of the American football player on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation in the sport, acknowledging that this can vary substantially among the unique positions. Outside of critical psychological aspects and team chemistry components, which are difficult to quantify, the pie chart shows the general relative importance of skills (30%), tactical awareness (50%), and physical capacities (20%) for American football success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

Targets of Physical Performance of American Football

The position-specific tasks required of AmF players (figure 20.1) tend to determine the physical characteristics they possess (22). Pincivero and Bompa (30) described four basic groups of established players as follows: (1) offensive and defensive linemen, (2) defensive backs and wide receivers, (3) linebackers, running backs, and tight ends, and (4) quarterbacks, punters, and kickers. Regardless of playing position, the predominant energy system of importance for all players is the anaerobic system (14). However, high-intensity interval training (HIIT) work and relief periods as determined should still be prescribed as specifically as possible for the varying positional groups that follow.

Offensive and Defensive Linemen

Typically the linemen, both offensive and defensive, are the largest athletes on the field. Players in this position exhibit higher percentages of body fat (OL, $25.3\% \pm 4.4\%$; DL $20.9\% \pm 6.5\%$) over most other positions and may weigh up to 140 kg (24, 31). The linemen positions must repeatedly apply explosive power efforts over very short distances. These athletes are characteristically stronger and more powerful than the other positions on the field, as they must possess the ability to be able to duel against opponents of similar size mirroring their position on the field (9, 30) (figure 20.1). While linemen display more physical contact with opposition players via specific blocking assignments and tackles (~ 4.4 per game (29)), they must nevertheless cover 200 to 600 yd per game (~ 7 -14 yd per play; ~ 38 -53 plays per game) including accelerations, decelerations, and changes of direction. This high-intensity work schedule in AmF players implies a high use of phosphocreatine and anaerobic glycolytic energy (15, 30). However, when such high-intensity efforts are repeated, this places additional demands on the aerobic system (see chapters 3 and 5), which are often less developed in AmF players (10, 14).

Defensive Backs and Wide Receivers

Defensive backs and wide receivers (figure 20.1) also share comparable body composition profiles. However, Bosch et al. (1) showed that WRs were slightly taller (WR 185.7 ± 3.9 cm; DB 182.2 ± 3.1 cm), slightly

heavier (WR 94.0 ± 6.0 kg; DB 90.8 ± 6.1 kg), and of a similar percentage body fat (WR $12.5\% \pm 3.1\%$; DB $12.1\% \pm 3.3\%$) as DBs. These players display the greatest locomotive activity demands in AmF (39), completing the greatest volume of running (~ 1200 yd per game; ~ 24 yd per play; ~ 43 plays per game), with much of this being high-speed running (~ 16 mph). These athletes require similar energy system contributions as others on the field, with slightly more reliance on the aerobic system due to the greater active distances traveled. To meet these demands, HIIT prescription includes short intervals, repeated-sprint training (RST), sprint interval training (SIT), and game-based HIIT (GBHIIT) (see below). Tactically, WRs and DBs have more space to run, which means they reach higher peak velocities and run longer distances in a game. Hard changes of direction, accelerations, and repeated longer high-intensity efforts, implying high neuromuscular and musculoskeletal demands, are required components of performance related to success in this positional group.

Linebackers, Running Backs, and Tight Ends

An AmF team's defense is often dictated by the success of its linebackers. They are high-velocity athletes balanced with strength and skill. Reading the play and being able to respond immediately to execute a tackle is the mainstay requirement of this position. Further, linebackers provide an extra run or pass protection option.

The principal ball carrier in the offense is known as the running back (halfback or fullback; figure 20.1). He is required to receive the handoff from the quarterback for a rushing play while also being needed to catch passes and block the opposition. The successful running back can use his agility to receive handoffs from the QB to attempt a rushing play. Running backs sustain the most severe ($>10G$ force) impacts during a match, whereas LBs endure lighter impacts (5.0 - $6.0G$ force) through collisions and tackles (39). Tight ends often are used as hybrid players who can play roles similar to those of offensive linemen, wide receivers, and running backs. They play next to the offensive linemen (figure 20.1) and often are required to block opponents on running plays. However, the TE may catch passes and be required to run the field. Players in all positions typically present with large statures (LB 109.9 ± 4.6 kg; RB 105.4 ± 8.5 kg; TE 113.9 ± 4.2 kg) and moderate percentage body fat ($\sim 16\%$ - 17%) (1). In addition, in order to accelerate and

decelerate repeatedly throughout a game, TEs often must push against opponents of similar size. All three positions complete ~550 yd of high-intensity running (~16 mph) in a game (~21-24 plays per game). Accordingly, these players require exposure to RHIE that involves accelerations so they are prepared for worst-case scenarios that occur during extended game play.

Quarterback, Punters, and Kickers

When we imagine AmF, we immediately think of the quarterback, who outside the coaching staff is the leader of the team. Quarterbacks typically possess lower values of body fat and high levels of speed and agility. They require good relative upper-body explosive muscular power to repeatedly throw the football downfield, targeting teammates with high precision. Despite these necessary attributes, the main feature of the QB's performance stems from a difficult-to-quantify game intelligence, tactical prowess, and ability to read the game, making decisions with split-second reaction time. The QB may throw ~36.3 times in a game (29) and cover ~550 yd per game over 47.9 plays (7.4 yd per play), relying on short, sharp multidirectional movements, accelerations, and decelerations. While the QB controls the offensive play of the game, he still requires enough strength and fat-free soft tissue mass to withstand regular collisions and impacts. QBs perform the majority of their training using individual short-interval HIIT, alongside their lead role in GBHIIT scenarios. Such training is considered pivotal to a QB's conditioning, alongside his copious training volume spent learning and refining skills.

The role of punters and kickers is simple: kick the football to score extra points, kick the ball off a placeholder to start play, or kick the ball in play to advance it down the field. The positions are highly specialized, but teams sometimes have the same person performing both roles. The kickers may need to kick or punt the football several times per game (punter, 11.6 per game; kicker, 9.4 per game) (29) depending on the success of the offense and the tactical context of the game, and cover up to 20 yd per play. Punters and kickers do not usually take contact, nor are they typically involved in tackling or core events in the general game. Although slightly lighter than QBs (K/P 98.4 ± 5.6 kg; QB 103.6 ± 13.9 kg), kickers and punters have similar proportions of body fat (K/P $19.2\% \pm 4.5\%$; QB $19.6\% \pm 4.6\%$). Punters and kickers are specialized

positions that require high levels of skill. Due to the tactical nature of their positions, they typically endure a very low level of locomotor activity in games. Accordingly, training programs for these positions use predominantly short-interval HIIT and RST plus their involvement in GBHIIT. In addition, their role sees them performing repeated kicking scenarios under pressure to improve accuracy.

Key Weapons, Manipulations, and Surveillance Tools

The weapons we use in AmF refer to the HIIT formats applied in our programs. These weapons are chosen to ensure we are targeting the desired physiological responses and physical adaptations, while our surveillance ensures we are meeting our directives. In this section, we present the different HIIT weapons we have used in preparing professional AmF players along with their manipulations and surveillance approaches for monitoring training effectiveness.

HIIT Weapons

The general breakdown of the different HIIT weapons we use throughout the annual season in AmF are shown in figure 20.3 and include GBHIIT or small-sided games (SSGs; ~60%), short-interval HIIT (~20%), RST (~15%), and SIT (~5%). These HIIT formats and their manipulations are described below.

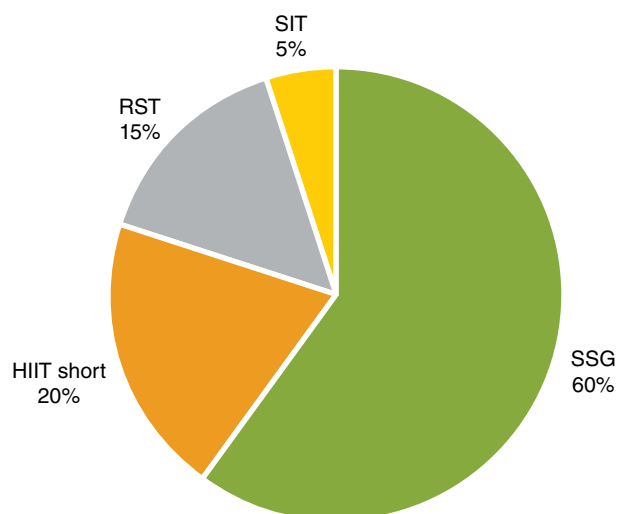


Figure 20.3 Percentage of the different HIIT formats (weapons) used throughout the annual season in American football: game-based HIIT or small-sided games (SSGs; ~60%), HIIT short intervals (~20%), RST (~15%), and SIT (~5%).

Manipulations of Interval Training Variables

Positions across the offensive and defensive roles in AmF are comparable to one another from an energetic standpoint, as they naturally mirror the movement demands. As such, training across offensive and defensive players is similar. Therefore, careful manipulation of the HIIT type, duration, intensity, and rest periods of each particular weapon are made within the weekly game cycle depending on team requirements.

Game-Based Training

While this book is about HIIT, generally defined as exercises eliciting close-to-maximal metabolic responses, for a comprehensive view of our AmF programming approach, we chose to include presentation of training from a slightly wider range of exercise intensities, including exercises performed at an intensity slightly below HIIT-specific intensities, as occurs with AmF-specific SSGs.

American football has a large training focus involving either the defensive group or the offensive group in which plays are simulated in a full and modified game format. This SSGs approach is traditional in college and professional teams and the main HIIT weapon used to build a player's conditioning. Therefore, teams will practice in their specific positional setups to optimize synchronicity of timing and strategy of plays. The combination of high-intensity running combined with specific skill involvement is seen as a favorable combination for AmF coaches over traditional running to condition the playing group. Unlike other football codes, namely soccer and Australian football in which running is more free-flowing, AmF practice sessions involve short explosive intervals with a high number of collisions. More recently, rules have been introduced by leagues to limit the number of sessions that can be completed with contact to mitigate risk of head injuries.

The breakdown of practice sees the offensive positions complete more locomotor activity by way of high-intensity running, using full-sized fields to invoke higher neuromuscular demands. This field-side modification helps to provide repeated bouts of work with increased mechanical work from accelerations, decelerations, and harder changes of direction (COD). This 5-a-side format (reduced from 11 in match play) with spatial constraints and noncontact SSG rules allows for short periods (~3 min) of play

with often multiple repetitions (~6-10). This format meets the longer-duration work requirements without overloading high-intensity running volume, while hitting type 4 targets.

HIIT With Short Intervals

Due to the wide variation of physiological demands across the various positions in AmF, a key weapon used by all positions naturally becomes HIIT with short intervals. Careful prescription of short intervals allows us to hit type 1, 2, 3, or 4 HIIT targets. Prescription of slightly longer intervals (≥ 15 s) in the off-season period are used to maintain specific components of the athlete's conditioning, as pre-season witnesses athletes resuming game-style play very quickly and they cannot afford to detrain. By managing the volume of this format, intensity of locomotor load, the neuromuscular fatigue, and anaerobic contribution can be manipulated accordingly, depending on the number of sessions planned for a day. This can be a useful format for rehabilitation in players who require lower neuromuscular loading (type 1) in the early phase before progressing through to type 2 and type 3 targets in preparation for type 4 short-interval work demands, involving COD running with hard decelerations. Rehabilitation players can build up to volumes of work required as percentages of their game demands calculated by known game run volume and intensity requirements. As deemed appropriate, players may complete 50% to 75% to 90% of the positional equivalent, with type 4 loading carefully controlled through prescription.

During the in-season period, there is a need for players to meet certain volumes of high-intensity running each week to lower injury risk and maintain fitness. Depending on how a game evolves, some players will not meet their demands. This may apply to both the main players and reserves, as there is no second tier competition from which players can get a game-like conditioning training load. For this situation, type 2 or 4 high-intensity running with high-grade agility and hard decelerations are required, which may serve to protect the hamstrings and mitigate deconditioning while maintaining neuromuscular loading. Such training ensures that reserve players are meeting the same game and training running loads as their senior counterparts. This training is also important for the backup kickers and punters and QB positions, who require kicking and throwing loads to match those of their game-playing counterparts.

Repeated-Sprint Training

Repeated-sprint training is a much used HIIT weapon in AmF players. The game demands of multiple positions require consistent all-out efforts that include COD and explosive efforts with short relief periods. This training replicates the intensity and frequency of game play in AmF. RST is used during both the late preseason and in the middle of the week during the in-season phase. It is commonly used in drills in which a player is required to hit tackle bags, run all out in a straight line, before changing direction in response to a stimulus, and then run again. Players may have multiple CODs requiring high levels of deceleration before applying an explosive effort again. These efforts are short (4-6 s) and based on their positional requirement. A typical format used for many positions might involve three series of 3 to 5 sprints with ~20 s of recovery over 5 to 6 min, which hits type 4 targets.

Sprint Interval Training

For the specialist positions who need to run the ball at maximum velocity down the length of the field in an attempt to score a touchdown, the SIT weapon naturally is employed as part of the periodized plan. The athletes complete this training individually when they are fresh and away from football-specific practices. The objective of this specialized session with select athletes is to be able to hit maximum velocity over ~10 repetitions over approximately 25 yd (similar to distances covered in a game per play for many positions) with extended relief periods (~2-3 min). This training forms a part of the late preseason and in-season phases and requires careful planning to prevent possible interference with other important training components and their associated neuromuscular loads (4).

Load Surveillance and Monitoring Tools

Microtechnology and the monitoring of player health have taken a longer time to gain traction in U.S. team sports, and as a result it is typically uncommon practice to see wearable technology being used in elite AmF. Over the last 5 yr, with the influence of European/UK and Australian sport scientists, both college and professional AmF have seen major advancements in this domain. The addition of complementary sport science to the area of strength and conditioning has led to the progression of understanding into the physiological demands and

responses associated with AmF (16, 17, 36). While tracking and monitoring systems still must undergo external validation, recent developments in GPS have seen embedded inertial sensors, which may be used to help understand collisions and positional differences in practice (38-41).

As discussed in chapter 8, collection of the session rating of perceived exertion (s-RPE) is a useful means to measure internal training load (13, 19), while high-definition camera systems can be useful for tracking external loads in training and games. Despite the need for validation, these systems together may be useful for quantifying overall workload to assist with alterations in weekly load distribution. This becomes important when the weekly microcycle is shortened from 7 d to 5 d during the competition phase due to scheduling (i.e., Monday night football prior to a Saturday afternoon game later in the week). Camera system limitations do arise, however, when the team is required to play an away game and train at remote locations beforehand (~6-8 games per season). Without consistent monitoring in these situations, session tracking is compromised (see related example in chapter 9). Heart rate is often monitored in non-football-related sessions, as well as in the rehabilitation setting to ensure the desired internal training load is adhered to. Such sessions may include stationary cycling, boxing, and HIIT running sessions, which are all common activities used throughout the rehabilitation phase.

Training Status Surveillance and Monitoring Tools

The training monitoring surveillance systems used in AmF are restricted to noninvasive measures to ensure compliance. With a squad that can be as high as 90 athletes in the preseason, players have many demands placed on them. For example, the preseason witnesses multiple overlaps in both position-specific and team sessions. Collectively with weight training, individual need workouts (HIIT doses for specific groups), alongside top-up conditioning for reserve players who may not have been exposed to sufficient game-time repetitions, monitoring systems must be simple, quick, and seamless for both athletes and practitioners.

The preseason period in AmF represents a discovery period for us, when we can debate and put forth what data we believe will be meaningful to collect, as surprisingly there are few longitudinal variables previously reported in either college or

professional teams (21, 37). Indeed, the desire to borrow systems from other sports that have monitored athletes in other football codes (namely soccer, rugby, and Australian football; see related chapters) is tempting, but practitioners must appreciate that the embryonic stage of buy-in for any elite sport can be difficult and slow. Best practice in such a sport may be to introduce only a few systems initially, while ensuring the data is collected well, before eventually becoming a consistent aspect of the team environment. Further establishment of the data analysis must then be meaningful enough to aid in the decision-making process before introducing any other noninvasive measurements deemed necessary. Collectively, GPS and accelerometry-derived data (including both locomotor and impacts), HR, s-RPE, countermovement jump scores for neuromuscular fatigue (6), alongside simple wellness questionnaires today may represent best practice scenarios for understanding time course changes of recovery and

response to training load (12, 27) (see also chapter 9). In order to aid player compliance, many of these systems can be integrated into the program and standardized in order to change the perception that an athlete is being tested or evaluated. The use of visual analogue scales for specific issues and to aid in understanding player readiness during periods of return to sport (28) complement the other systems outlined. An example of the typical weekly team training loads experienced throughout a season are shown in figure 20.4.

Strategies for Structuring the Training Program Using HIIT

Strategies for structuring HIIT programming in AmF may be dictated partially by several controllable and uncontrollable factors, before proper progression and periodization can be appropriately implemented. This section discusses some of the

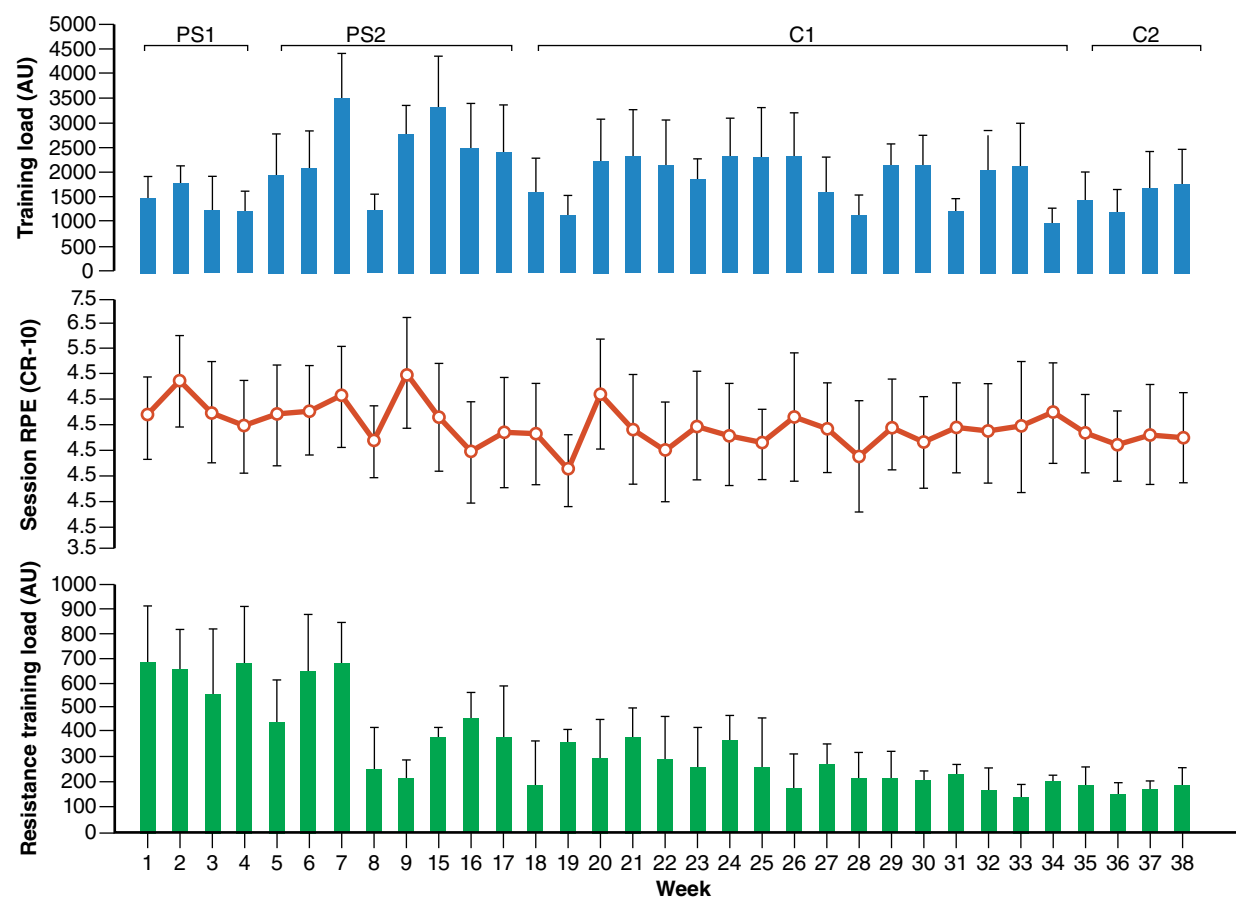


Figure 20.4 Example of seasonal weekly team training loads (AU) and session-RPE and resistance training loads (AU) during a professional American football league season. PS1: preseason phase 1; PS2: preseason phase 2; C1: competition season; C2: finals.

nuances of training program and HIIT structuring in AmF.

Controllable and Uncontrollable Factors

The uncontrollable factor in AmF at the professional level is its schedule, which determines game day, venue, and associated interstate travel (including up to three time zones in either easterly or westerly direction). The practice period, which has strict league rules regarding the amount of time players are accessible and regulations concerning the number of sessions during which contact time is permitted, is set prior to the start of the season. These aspects dictate how the program is structured in the weekly cycle and, most importantly, in accordance with the head coach's philosophies. Players and coaches are also required to partake in extensive media commitments throughout the week, both in preseason and in-season.

The planning of the preseason is carefully constructed around the technical and tactical philosophies of the head coach. For this, each group of positions outlined tends to separate and focus on individual positional roles. Indeed, practice sessions for these groups are often relatively long due to the high number of plays required to orchestrate their function without error. Therefore, completing the daily practice plans in blocks that ensure yards per minute are maintained without diluting team practice due to fundamental teaching components that may be lacking in particular athletes is a necessity. A predetermined plan of training intensity and duration is set early (chapter 8) so that drills can be established noting player availability for contact and noncontact aspects. Once the initial technical and instructional blocks are achieved, the inclusion of different HIIT components is usually blended into further tactical sessions. With live monitoring of GPS (high-intensity running and volume of running), performance staff can ensure predicted physical requirements are met. Often starters will receive appropriate training doses that include physical game outcomes, however, nonstarters will complete complementary training sessions that use additional specific HIIT work to ensure a matched training load. This load often exceeds that of consistent starters.

Program Structure and Progression

Training for professional AmF has evolved in recent years so that today it is more specific to the

requirements of the individualized positions. Teams typically do not follow any specific periodized model in terms of varying training cycles during the in-season period (figure 20.4). The flexible nonlinear programming approach offers variation in training to ensure more responsive changes in the athlete (23). The off-season period begins after the Super Bowl (in February) and ends with the beginning of the next season's practices in late June, running until late July, followed by a 2 wk preseason period with lifting sessions, meetings, and limited practices. The next period of 6 wk features a weekly game factored into the cycle and progressive changes in the volume and intensity of the program. Upon completion of this phase, players ready themselves for the in-season phase in which they compete in 16 games over a 17 wk period. If successful during the regular season, the opportunity to commence post-season games may see teams play in three final high-profile playoff games.

Different aspects of the annual season can be challenging for players and performance staff. The off-season provides the biggest challenge to staff due to the inability to track and monitor players while they are away from the home training facilities. Programs are issued and many athletes complete these, but compliance is never 100%; success depends on the individual athlete's motivation. To ensure players return to the preseason in an acceptable level of physical condition, many targets are provided to players beforehand to help them avoid detraining. An adequate off-season is nevertheless important, and allows players to enter preseason with a lower risk of injury. Players must be able to perform a high volume of high-intensity work, involving slightly longer running intervals (HIIT with short intervals targeting type 4), multiple weight sessions, team technical and tactical practices without contact, and off-legs conditioning work. Work progresses carefully and involves a higher volume of straight-line running and lower-grade agility. Players involved in kicking and throwing progress in terms of intensity and volume of these activities, and the placement of these in the program is critical. There is always a desire for coaching staff to attain higher repetitions of both kicking and throwing, but this is where monitoring is essential to achieve controlled and balanced loading alongside the conditioning component of the program to ensure the incidence of injury is mitigated. Figure 20.5 shows an example of the weekly running loads experienced throughout a typical AmF season.

Preseason games start very early, and the use of wearable technology for monitoring load is still

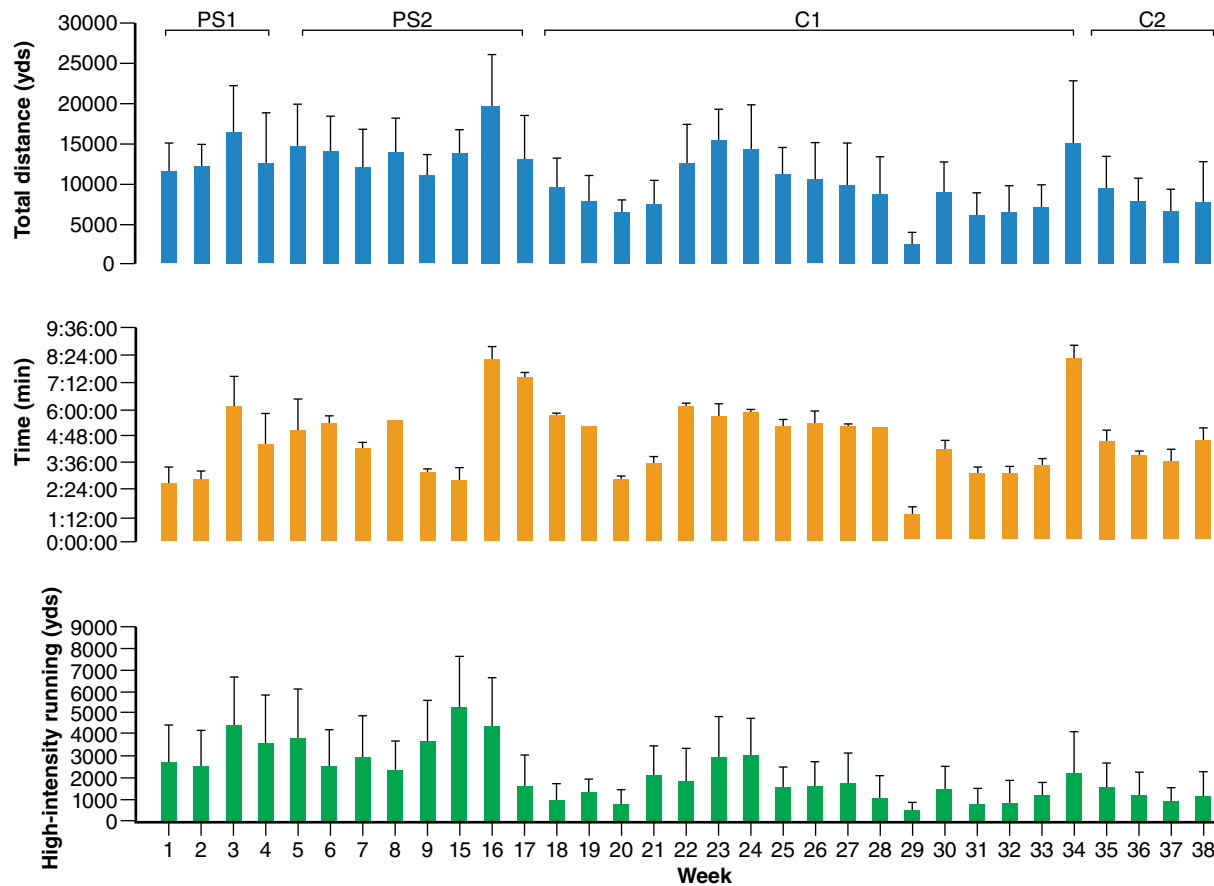


Figure 20.5 Example of seasonal weekly team running loads, total distance (yards), time (minutes), and high-intensity running (16 mph) yards during a professional American football league season. PS1: preseason phase 1; PS2: preseason phase 2; C1: competition season; C2: finals.

limited in this regard. That being said, players tend to increase RHIE and reduce rest intervals progressively, while increasing the number of harder accelerations and decelerations to replicate position-specific game-like situations. The efforts are short (4-6 s) and based on positional requirements. A typical example of training across many positions might involve three series of 3 to 5 sprints with ~20 s recovery over 5 to 6 min in which type 4 targets are reached. The priority focus is always on game 1 of the regular season, so the need to gain strength and muscle mass remains a primary focus throughout this period.

The in-season period is often the easiest phase of training to manage due to weekly training loads remaining constant over the 17 wk period to ensure optimal performance occurs at game time. Similar to most weekly football codes, there is a period of recovery that follows from the damage incurred from the weekend game (16, 17) before a buildup to the biweekly technical and tactical practices, followed

by unloading for recovery for the game. Importantly, ensuring an appropriate dose of quality high-speed running is met not only maintains fitness characteristics but also protects from injury (11, 26, 34).

Incorporation of HIIT

In AmF, planning and sequencing of HIIT between games is based on a few important aspects: the amount of work completed by the player, the level of soreness and fatigue, and, most importantly, the coach's decision. Often after a game, the coach will want to work on technical aspects to improve player and team performance. This often takes priority over most other decisions, desires, and needs of other staff. The starters and nonstarters provide an interesting scenario in this regard, as they have different loading volumes postgame. Nonstarting players are required to top up their training load as early as possible, so they follow a similar time course of training load response and recovery adaptation as their starting counterparts.

Training sessions in AmF were traditionally completed using number of repetitions or plays. For example, coaches used to use language such as “We will be running 80 plays today” instead of “We need to complete 4000 yd, 75% of which will include high-intensity running.” So the shift in programming due to the inclusion of GPS/RFID tracking has markedly changed top-up session planning and execution. While this ideal scenario is now often the case, coaches still may revert back to use of more traditional language.

The higher-velocity running, sharp changes of direction, as well as accelerations and decelerations are avoided during the first 2 d after a game. We also consider the demands of other training in which HIIT is incorporated as well as the loading patterns within the weekly cycle, as these factors can influence the metabolic and neuromuscular responses to the HIIT (4, 5). The programming emphasis relies on HIIT incorporated into the main training session prior to technical and tactical training in the preseason, as

the emphasis from the coach is to have players perform skills under slight fatigue. There is another portion in the form of SSG often implemented at the mid to latter end of the coach’s session. The aim here is to have positional teams focus on their specific plays. The offensive and defensive linemen, for example, will complete multiple short (<3 s) efforts that involve pushing opponents off after hard, short accelerations. Ideally, the plan would be to incorporate HIIT weapons that hit type 1 and type 3 targets, to avoid the neuromuscular loading and muscle damage incurred from the full-contact game, but depending on the outcome (good or bad), there can often be running that occurs, albeit undesirable in the eyes of the performance staff.

Sample Training Programs

An example of a weekly training program structure for professional AmF for preseason and in-season phases is shown in tables 20.1 and 20.2, respectively.

Table 20.1 Sample Preseason Training Schedule

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
AM	Off	Rehab running group	Rehab running group	Rehab running group	Rehab running group	Rehab running group	Rehab running group
		Team meeting Treatment physiotherapy Weights for all groups	Team meeting Treatment physiotherapy	Off	Team meeting	Team meeting Treatment physiotherapy Weights, individual needs	Team meeting Treatment physiotherapy Weights, individual needs
		Line meeting Practice, tactical session	Line meeting HIIT running Practice, main session		Line meeting HIIT running types 1, 2 Practice, main session	Line meeting Practice, main session HIIT running types 1, 2, 3 (position specific)	Practice, tactical session Individual off-legs conditioning
PM		HIIT running types 1, 2 Treatment physiotherapy	Weights for all groups		Weights for all groups Treatment physiotherapy	Weights for all groups Treatment physiotherapy	Weights for all groups Treatment physiotherapy

Table 20.2 Sample Competition Season Training Schedule

	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
AM	Rehab running group	Rehab running group	Rehab running group	Rehab running group	Rehab running group	Rehab running group	Rehab running group
	Team meeting Weights for rehab group	Team meeting Treatment physiotherapy Weights for all groups	Team meeting Treatment physiotherapy		Team meeting Specialist HIIT running group types 1, 2, 3, 4 (position specific)	Team meeting Weights, individual needs	Team meeting Weights, individual needs
	Line meeting	Line meeting	Line meeting HIIT running types 2, 4 (position specific) Practice, main session	Off	Line meeting HIIT running Practice, main session	Line meeting Practice, walk through HIIT running type 4 (position specific)	Practice, walk through Individual off-legs conditioning
PM	Game	Warm up individual top-up HIIT types 1, 2 Individual off-legs conditioning Tactical overview session review Treatment physiotherapy	Weights for all groups	Treatment physiotherapy Review meeting	Treatment physiotherapy	Treatment physiotherapy Review meeting	Treatment physiotherapy

CASE STUDY

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This story describes how we learned to monitor and assess training more precisely and how new aspects being introduced into a high-performance culture should be well understood prior to their implementation. This is very much a “lessons learned” experience, in which the timely processing of information and understanding specificity came into play.

The start of preseason saw a new staff member added to the performance department we were part of. This skill acquisition expert was to provide the link between the coaches and the performance department. The position was to use game-based training to better prepare the players for the specific game demands. While the current staff were confident this was already being achieved, this was an upper-management hire and not one vetted by the high-performance director (HPD). At the same time, we were in the embryonic stage of microtechnology use in American football. While the technology was familiar to our sport science staff, its practical use and understanding of useful metrics were still undergoing preliminary in-house reliability and validity evaluation (see chapter 9).

During the second half of preseason, the new skill acquisition staff member was granted permission to monitor the athletes and incorporate the game-based training component of the training sessions. The coach was convinced that this was the direction to go and disregarded some of the more traditional run-based HIIT sessions in which specific loading was planned using run distances and rest intervals. Even without the microtechnology, we could still set a course and time our rest intervals to ensure prescribed and actual doses were aligned. The conditioning coach, HPD, and sport scientist observed the transition to the game-based training drills and noticed the amount of teaching required during stoppage in training. The performance group became quite frustrated when the skill acquisition staff member informed us of an issue with the GPS units and that he would get information through to the performance group when he could address the technical downloading issues. This seemed to us merely a ploy to avoid presenting his information and controlling the narrative.

Frustration grew among the performance staff and the control of training dose was lost. It was now difficult to assess the players’ load being completed, and staff highlighted this issue in several subsequent meetings. But when a coach is sold that a new protocol is the right way to do things, it can be hard to sway him back. It’s important to emphasize here that the concept of game-based training was not new to the current performance staff, with the sport scientist at the club just completing student supervision of a PhD in rugby-based SSG, and therefore was well-versed in how to incorporate SSG in practice. Of course, these should form part of training, and so long as they are monitored properly, can be utilized as needed (see chapter 5). Needless to say, the inability to get direct answers to the monitoring process and loading left the performance staff extremely frustrated.

Fast forward to the start of the competition season. Game 1 sees calf tears in two key starters. Game 2 sees another calf tear and a hamstring strain. Games 3 and 4 witness yet another calf tear in each game. By game 5, we had an unofficial “calf crisis” when we strained our sixth calf. Six starters were out with similar injuries. The heat was turned up on the HPD and medical department, who were questioned regarding their process. The HPD called for an external performance audit of the department to get an independent answer on the so-called “calf crisis.”

In the meantime, the conditioning coach and sport scientist collected all the GPS units and conducted their own investigation. What they discovered was that many of the sessions had either not been downloaded or had not been processed. Upon completion of their report, they discovered that the early competition games were run at 157 to 164 yd per minute. The game was fast! The analysis of the preseason sessions conducted by the skill acquisition expert were shown to be high in running volume, but with the constant coaching

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CASE STUDY *(CONTINUED)*

stoppage and suggested spatial constraints implemented, the sessions achieved only 98 to 101 yd per minute. The in-season sessions were also of low intensity and similar volume. The new stimulus of the weekly game, however, resulted in a huge spike in yards per minute and high-intensity running. The players were not adequately prepared for these increases in intensity and high-speed running. Despite there being some peaks in work rate during some drills, the rest intervals during the game-based training were too long and the intensity not maintained. Much of the work was completed at subpar levels that did not align with the game demands.

Of course, now we were in-season, when a change in preparation is difficult. Thus, solutions required exceptionally controlled running progressions and management.

To summarize, a number of our players had detrained because the principles of training as you play were not met. Data was not processed in a timely fashion, so adjustments could not be made for subsequent sessions. This occurrence affected the season drastically, as player management was extremely difficult. The steady implementation of appropriate volumes of high-speed running helped mitigate some of the issues going forward, allowing us to survive the season. But we learned a great lesson. The need to ensure the right staff complements a department, alongside an understanding of how any planned training should be measured to be managed, is a must in elite sport.